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INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Employee Management System Using SQL

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Batch No : 520

About me

- **Background : Bachelor of Technology (B.Tech)** in **Computer Science Engineering(CSE)** from Mahatma Gandhi Institute of Technology(MGIT)

- **Why you want to learn Data Analyst:** I want to learn Data Analytics because it enables me to transform data into meaningful insights that support better business decisions. I am interested in analyzing data, identifying patterns, and solving real-world problems. My goal is to build a successful career as a Data Analyst and continuously enhance my analytical and technical skills.

- **work experience :** I have 2 years of Professional experience in LTM

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Agenda

- Introduction
- Business Problem Statement
- Objective of the Project
- Database Tables
- Key analysis questions (use cases)
- Conclusion (Key finding overall)
- Experience/Challenges

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Introduction

The Employee Management System is a SQL-based database project developed to organize and manage key employee-related information within an organization. The system integrates employee details, departments, job roles, salary and bonus structures, qualifications, leave records, and payroll into a structured relational database.

Using SQL, the project enables efficient data management, retrieval, and analysis, helping identify important patterns in workforce distribution, compensation, leave records, and payroll. Overall, the project demonstrates how a well-structured relational database can transform HR data into meaningful and actionable insights.

Business Problem Statement

Design and implement an Employee Management System that efficiently stores and manages employee-related data within an organization. The system needs to track various aspects of employee information, including personal details, job roles, salary structures, qualifications, leave records, and payroll data. The system should ensure the integrity and consistency of data by using relational tables with appropriate foreign keys and cascading actions.

The system should allow for easy management and querying of employee data, providing insights such as payroll calculation, leave tracking, and department-specific job roles. The goal is to streamline HR operations, ensuring that all relevant employee data is accessible and accurately updated across different modules.

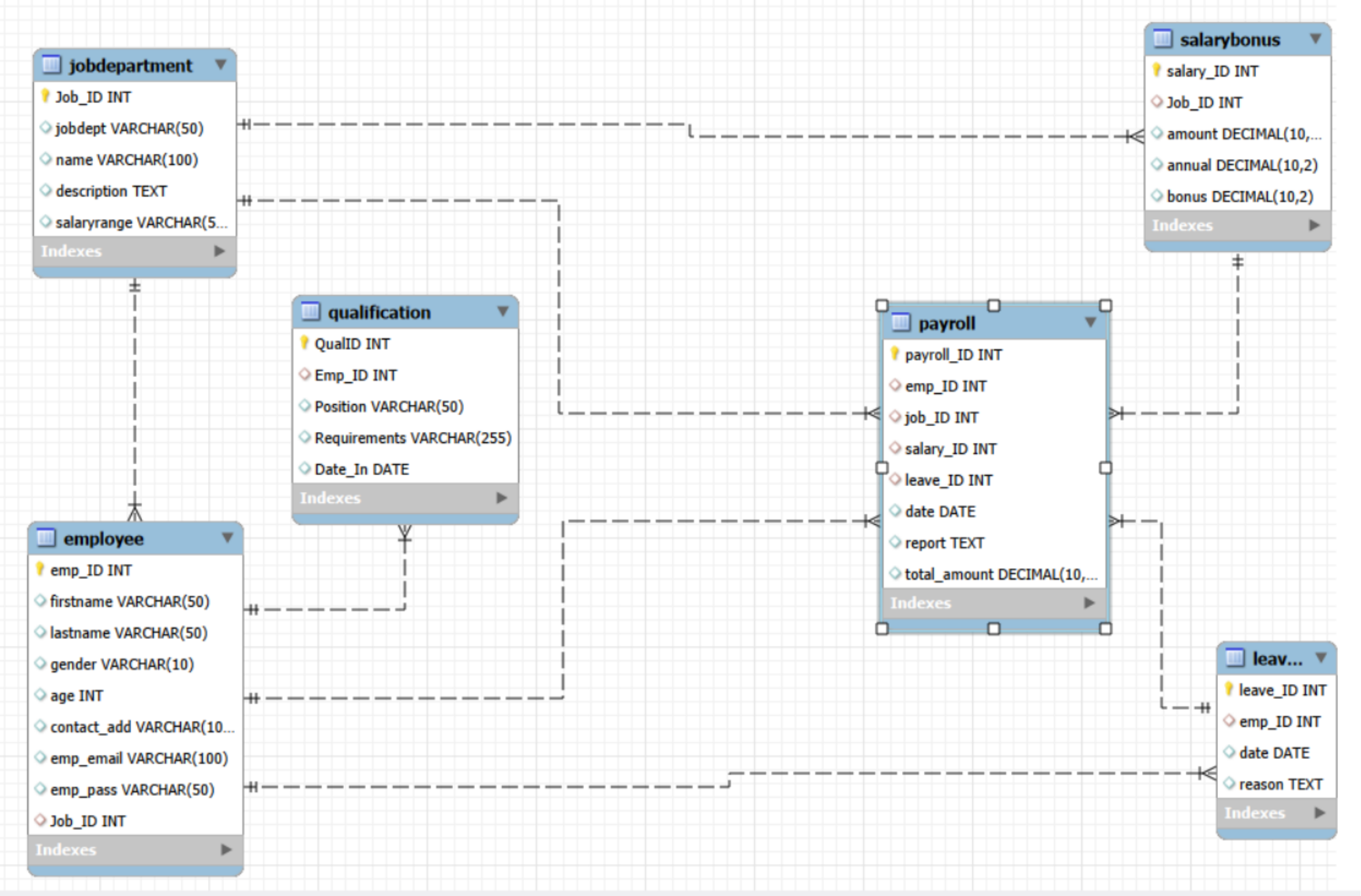
Objective

The objective of this project is to design and implement an Employee Management System using SQL that efficiently stores, manages, and analyzes employee-related data.

Key Objectives

- Manage employee and department information
- Track jobs, salaries, qualifications, leaves and payroll
- Establish relationships using Primary and Foreign Keys
- Maintain data integrity and consistency
- Perform SQL-based analysis to generate meaningful HR insights

ER Diagram



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Database Tables

1. JobDepartment: Stores department and job-role information. It also contains the salary range associated with each role.

- Fields : Job_ID, jobdept, name, description, salaryrange

2.SalaryBonus: Stores salary, annual salary and bonus information for each job role.

- Fields: salary_ID, Job_ID, amount, annual, bonus

3. Employee: Stores the main employee information, including personal details and the job assigned to the employee.

- Fields: emp_ID, firstname, lastname, gender, age, contact_add,emp_email,emp_pass,Job_ID

Database Tables

4. Qualification: Stores qualification information associated with employees, including position requirements and qualification date.

- Fileds : QualID, Emp_ID, Position, Requirements, Date_In

5.Leaves: Records employee leave information, including the leave date and reason.

- Fileds: leave_ID, emp_ID, date, reason

6.Payroll: Stores payroll records and connects employees with their job, salary and leave information.

- Fields: payroll_ID, emp_ID, job_ID, salary_ID, leave_ID, date, report,total_amount

Key Analysis Questions(Use Cases)

1. Which departments have the highest number of employees?
2. What is the average salary per department?
3. How many job roles exist in each department?
4. Which job roles offer the highest salary?
5. How many employees have at least one qualification listed?
6. Which positions require the most qualifications?
7. Which year had the most employees taking leave?
8. What is the average number of leave days taken by its employees per department?
9. What is the monthly payroll processed?
10. What is the average bonus given per department?

Employee Analysis

1. Which departments have the highest number of employees?

```sql
-- Which departments have the highest number of employees?
select jd.jobdept Department,count(e.emp_id) Total_Employees
from jobdepartment jd
join employee e
on jd.job_id = e.job_id
group by jd.jobdept
order by Total_Employees desc
limit 5;        -- top 5 departments, having highest number of employees
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fet

Department	Total_Employees
Finance	9
IT	9
Operations	8
Marketing	8
Engineering	7

Tables:

- JobDepartment
- Employee

Insights:

- Finance and IT have the highest number of employees, with 9 employees each.
- This means these two departments have the largest workforce in our dataset.

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Employee Analysis

2. What is the average salary per department?

```sql
-- What is the average salary per department?
select jd.jobdept Department,round(avg(sb.amount),2) Average_Salary
from jobdepartment jd
join salarybonus sb
on jd.job_id = sb.job_id
group by jd.jobdept
order by Average_Salary desc;
```

Result Grid Filter Rows: Export: Wrap Cell Content:

Department	Average_Salary
Legal	84600.00
Engineering	81142.86
Sales	75428.57
Finance	72333.33
IT	70888.89
Operations	68750.00
Marketing	65625.00
HR	62571.43

Tables:

- JobDepartment
- SalaryBonus

Insights:

- The average salary is different across departments.
- Legal has the highest average salary, which indicates that employees in this department have relatively higher salary levels.

Job Role & Department Analysis:

3.How many job roles exist in each department?

```sql
-- How many different job roles exist in each department?
select jobdept Department,count(distinct name) Total_Job_Roles
from jobdepartment
group by jobdept
order by total_job_roles desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Department	Total_Job_Roles
Finance	9
IT	8
Operations	8
Engineering	7
HR	7
Marketing	7
Sales	7
Legal	5

Tables:

- JobDepartment

Insights:

- The number of job roles varies from one department to another.
- This shows that some departments have a wider range of responsibilities and positions.

Job Role & Department Analysis:

4. Which job roles offer the highest salary?

```sql
-- Which job roles offer the highest salary?
select jd.name Role_Name,sb.amount Salary
from jobdepartment jd
join salarybonus sb
on jd.job_id = sb.job_id
order by sb.amount desc
limit 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Role_Name	Salary
Finance Director	170000.00
Engineering Director	160000.00
Marketing Director	150000.00
IT Director	150000.00
Sales Director	140000.00
Operations Director	135000.00
Senior Legal Consultant	130000.00
HR Director	120000.00
Senior IT Consultant	120000.00
Senior Engineering Manager	115000.00

Tables:
- JobDepartment
- SalaryBonus

Insights:
- Director-level roles have the highest salaries in the dataset.
- This shows a clear relationship between job seniority, responsibility and compensation.

Qualification & Skills Analysis:

5. How many employees have at least one qualification listed?

```sql
-- How many employees have at least one qualification listed?
select count(distinct emp_id) Qualified_Employees
from qualification;
```

Result Grid Filter Rows:

Qualified_Employees
60

Table:

- Qualification

Insights:

- All 60 employees have a qualification record.
- So, the current dataset has complete qualification information for the employees.
- HR can therefore use the qualification data for employee skill and capability analysis.

Qualification & Skills Analysis:

.*6.* Which positions require the most qualifications?

```sql
-- Which positions require the most qualifications?
select position,count(*) total_qualifications
from qualification
group by position
order by total_qualifications desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

position	total_qualifications
HR Assistant	1
HR Manager	1
IT Support	1
Software Engineer	1
Data Analyst	1
Sr. Data Analyst	1
Marketing Executive	1
Marketing Manager	1
Finance Analyst	1
Sr. Finance Officer	1
Legal Advisor	1
Office Admin	1
HR Executive	1
Network Admin	1
Frontend Developer	1
Backend Developer	1

Tables:

- Qualification

Insights:

- In the current dataset, each position has one qualification record.
- So, there isn't enough variation to say that one position requires more qualifications than another.

Leave & Absence Analysis:

7. Which year had the most employees taking leave?

```sql
-- Which year had the most employees taking leaves?
select year(date) year ,count(distinct emp_id) Employees_on_Leave
from leaves
group by year(date)
order by Employees_on_Leave desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

year	Employees_on_Leave
2024	60

Tables:

- leaves

Insights:

- 2024 is the only year available in the leave data.
- All 60 employees have a recorded leave entry for that year.
- Therefore, the current data is sufficient for 2024 leave tracking, but not for year-over-year trend analysis.

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Leave & Absence Analysis:

8. What is the average number of leave days taken by its employees per department?

```sql
-- What is the average number of leave days taken by its employees per department?
select jd.jobdept,round(count(l.leave_id)/count(distinct e.emp_id),2) Avg_leaves
from employee e
join jobdepartment jd
on e.job_id = jd.job_id
left join leaves l
on e.emp_id=l.emp_id
group by jd.jobdept;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

jobdept	Avg_leaves
Engineering	1.00
Finance	1.00
HR	1.00
IT	1.00
Legal	1.00
Marketing	1.00
Operations	1.00
Sales	1.00

Tables:

- Employee
- JobDepartment
- leaves

Insights:

- The average is approximately one leave record per employee across departments.
- So, there is no major difference in leave frequency between departments in the current dataset.

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Payroll & Compensation Analysis :

9. What is the monthly payroll processed?

```sql
-- What is the total monthly payroll processed?
select date_format(date,'%Y-%m') Month,sum(total_amount) Monthly_payroll
from payroll
group by Month
order by Month;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Month	Monthly_payroll
2024-04	2778000.00

Tables:

- Payroll

Insights:

- The company processed approximately ₹27.78 lakhs in payroll for April 2024.
- This represents the organization's monthly payroll requirement for the available period.
- Management can use this figure for payroll budgeting and financial planning.

Payroll & Compensation Analysis :

10. What is the average bonus given per department?

```sql
-- What is the average bonus given per department?
select jd.jobdept,round(avg(sb.bonus),2) Average_Bonus
from salarybonus sb
join jobdepartment jd
on sb.job_id=jd.job_id
group by jd.jobdept
order by Average_Bonus desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

jobdept	Average_Bonus
Legal	13300.00
Engineering	12571.43
Sales	11214.29
Finance	10666.67
IT	10444.44
Operations	9687.50
Marketing	9125.00
HR	8171.43

Tables:

- SalaryBonus
- JobDepartment

Insights:

- Legal has the highest average bonus among the departments.
- This indicates relatively higher average incentive allocation for employees in Legal.
- The insight can support bonus benchmarking and incentive planning across departments.

Insights

- 60 employees are currently recorded in the system.
- Finance and IT have the highest number of employees, with 9 employees each.
- Legal has the highest average salary, while senior/director-level roles receive the highest salaries.
- Finance has the highest total salary allocation of Rs.651K.
- All 60 employees have qualification records in the current dataset.
- The dataset contains 60 recorded leave instances, with one leave record per employee.
- Finance has the highest total bonus allocation of Rs.96K.
- April 2024 payroll processed was approximately Rs.27.78 lakhs.

Conclusion

The Employee Management System project was developed to organize and manage employee-related information in a structured relational database. The project connects employee details with departments, job roles, salaries, qualifications, leaves, and payroll, making the information easier to manage and retrieve. Using SQL queries, we analyzed different HR aspects and generated useful insights related to workforce distribution, compensation, qualifications, leave records, bonuses, and payroll. Overall, this project helped demonstrate how a relational database and SQL can be used to manage HR data efficiently and turn it into meaningful information for organizational decision-making.

Experience/Challenges

- Understanding how to connect the six tables using Primary Keys and Foreign Keys.
- Writing SQL queries involving multiple JOINs and aggregate functions.
- Understanding which table should be used for each business question.
- Working with salary, leave, qualification, and payroll data together.
- Handling date-based analysis such as year-wise leaves and month-wise payroll.
- Understanding the difference between employee-level and department-level analysis.
- Interpreting the query results and converting them into meaningful business insights.
- This project improved my confidence in SQL, database relationships, data analysis, and problem-solving.

THANK YOU

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